

Risk-Return Analysis and Portfolio Optimization



Master of Data Analytics, University of Niagara Falls

DAMO611 – Data Analytics Case Study 3

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Statement of Purpose

Problem:

how to wisely spread their money across different parts of the market

Goal:

Use historical stock data (Finance, Technology, Energy sectors) to analyze risk-return patterns & optimize portfolios.

Scope:

Data Collection, Understanding, Visualization phases, Data Modeling and Evaluation



Background Research

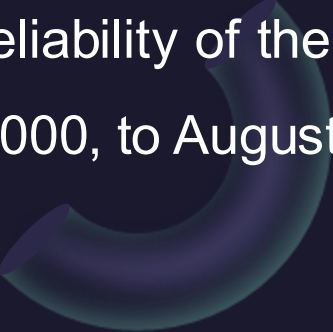
Modern Portfolio Theory (MPT):

- Pioneered by Harry Markowitz (1952), MPT revolutionized investment science by analyzing a portfolio's risk and return as a whole.
- **Sharpe Ratio:** Introduced by William F. Sharpe (1966), this metric measures a portfolio's performance by quantifying the excess return earned per unit of volatility. A higher Sharpe Ratio indicates a more efficient and desirable investment.

Ticker	Company Name	Sector
AAPL	Apple Inc.	Technology
AMZN	Amazon.com, Inc.	Technology
MSFT	Microsoft Corporation	Technology
NVDA	NVIDIA Corporation	Technology
GS	The Goldman Sachs Group, Inc.	Finance
JPM	JPMorgan Chase & Co.	Finance
MS	Morgan Stanley	Finance
COP	ConocoPhillips	Energy
CVX	Chevron Corporation	Energy
SLB	Schlumberger Limited	Energy
XOM	Exxon Mobil Corporation	Energy



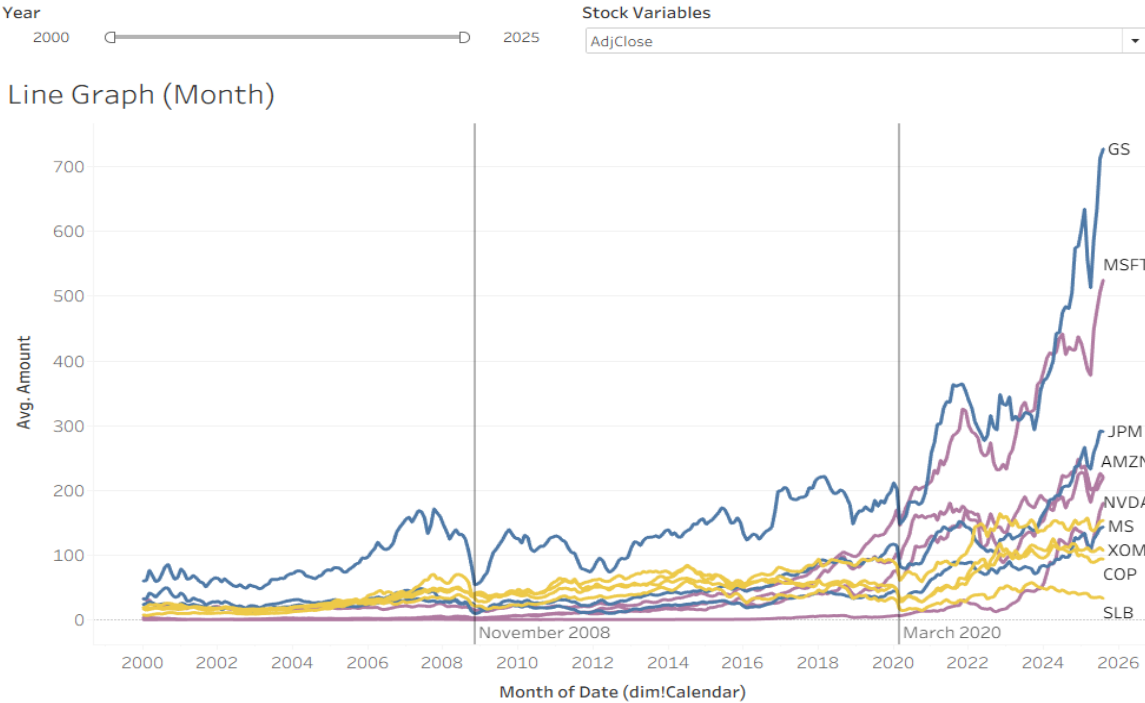
Data Collection & Understanding

- **Data Source:** The primary data source was the yfinance Python library, which provides access to historical and real-time data from Yahoo Finance.
 - **Data Fields:** The dataset includes key financial metrics for each company, such as Adjusted Close Price, Close Price, High Price, Low Price, Open Price, and Volume. The Adjusted Close Price was used as the primary basis for calculating daily returns.
 - **Data Quality:** The dataset is clean and complete, with no missing values, ensuring the reliability of the analysis. It includes 6,443 daily observations for 11 stocks from January 3, 2000, to August 14, 2025.
- 



Exploratory Analysis

Case Study 3 - Part 1 - Dashboard



Case Study 3 - Part 1 - Dashboard

Correlation Matrix - Adj Close

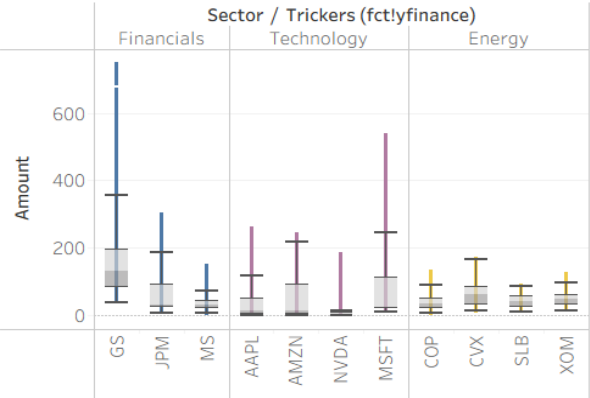
Year: 2000 ————— 2025

Correlation: -1.000 to 1.000

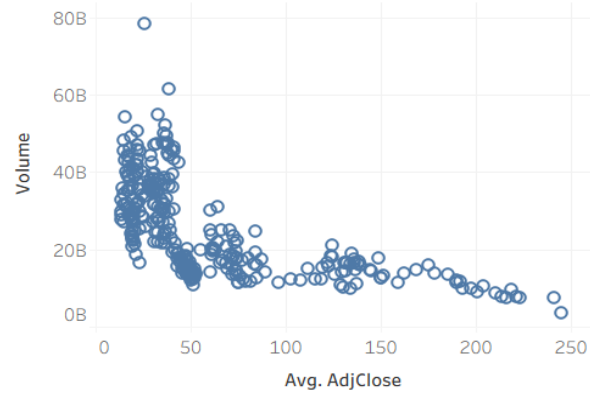
Sector: Financials (Blue), Technology (Purple), Energy (Yellow)

Ticker 2	AAPL	AMZN	COP	CVX	GS	Ticker 1	JPM	MS	MSFT	NVDA	SLB
AMZN	0.950										
COP	0.877	0.782									
CVX	0.882	0.830	0.966								
GS	0.930	0.899	0.852	0.861							
JPM	0.949	0.952	0.842	0.870	0.973						
MS	0.913	0.863	0.770	0.744	0.935	0.918					
MSFT	0.987	0.956	0.855	0.854	0.936	0.963	0.926				
NVDA	0.822	0.760	0.706	0.671	0.884	0.871	0.844	0.856			
SLB	0.003	-0.061	0.365	0.360	0.153	0.048	-0.077	-0.046	-0.010		
XOM	0.816	0.717	0.960	0.954	0.849	0.820	0.709	0.793	0.726	0.490	

Box Plot - Adj Close



Scatter Plot



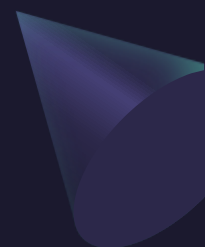


Hypothesis 1 (Sector Risk-Return Profile)

Hypothesis 1: Sector Risk-Return Profile

H0: There is no significant difference in the average returns and volatility across the Technology, Finance, and Energy sectors.

H1: There is a significant difference in the average returns and volatility across the Technology, Finance, and Energy sectors, with Technology having higher returns and volatility, while Finance and Energy have lower returns and volatility.



Hypothesis 1 (Sector Risk-Return Profile)

```
# Select only numeric columns (the tickers)
numeric_returns = daily_returns.select_dtypes(include='number')

# Average daily return per ticker
mean_daily = numeric_returns.mean()

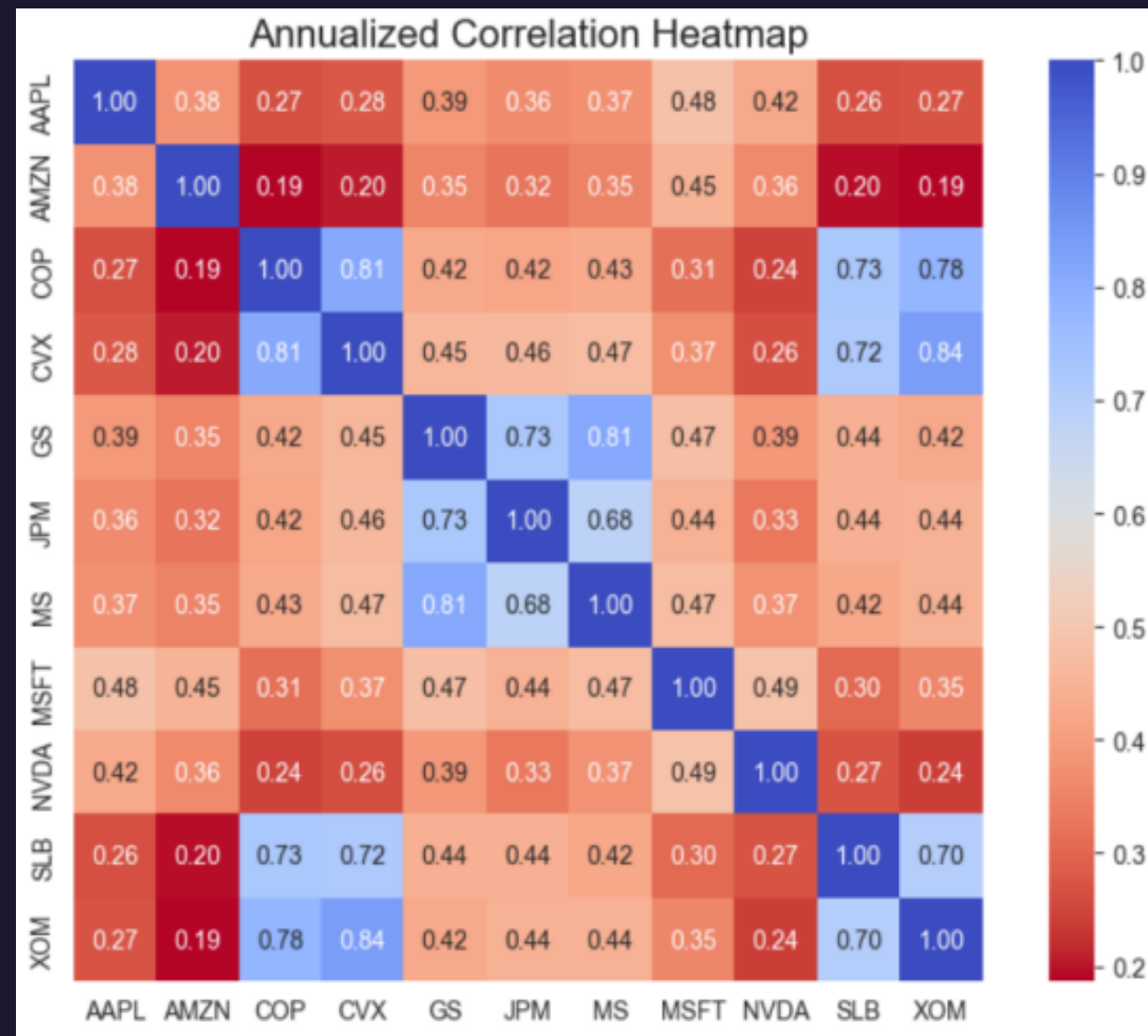
# Approximate annualized return (linear)
mean_annual_linear = mean_daily * 252

# Compound annualized return
mean_annual_compound = (1 + mean_daily) ** 252 - 1
```

```
=== Annualized Returns Sorted by Compound Return ===
      Linear Annualized   Compound Annualized
NVDA      0.470665      0.600355
AAPL      0.297493      0.346242
AMZN      0.272138      0.312575
JPM       0.168703      0.183702
MS        0.165045      0.179382
GS        0.161292      0.174967
COP       0.155610      0.168314
MSFT      0.148999      0.160621
CVX       0.127886      0.136387
XOM       0.108057      0.114086
SLB       0.104520      0.110154
```

Hypothesis 1 (Sector Risk-Return Profile)

- Overall, combining stocks from different sectors can reduce portfolio risk due to low intersector correlations
- Therefore, the null hypothesis (H0) of no significant difference can be rejected in favor of H1, which aligns with the sector risk-return profiles observed.



Hypothesis 2: Diversification Benefits

H0: A diversified portfolio across the three sectors does not have a significantly higher Sharpe Ratio than a single-sector portfolio.

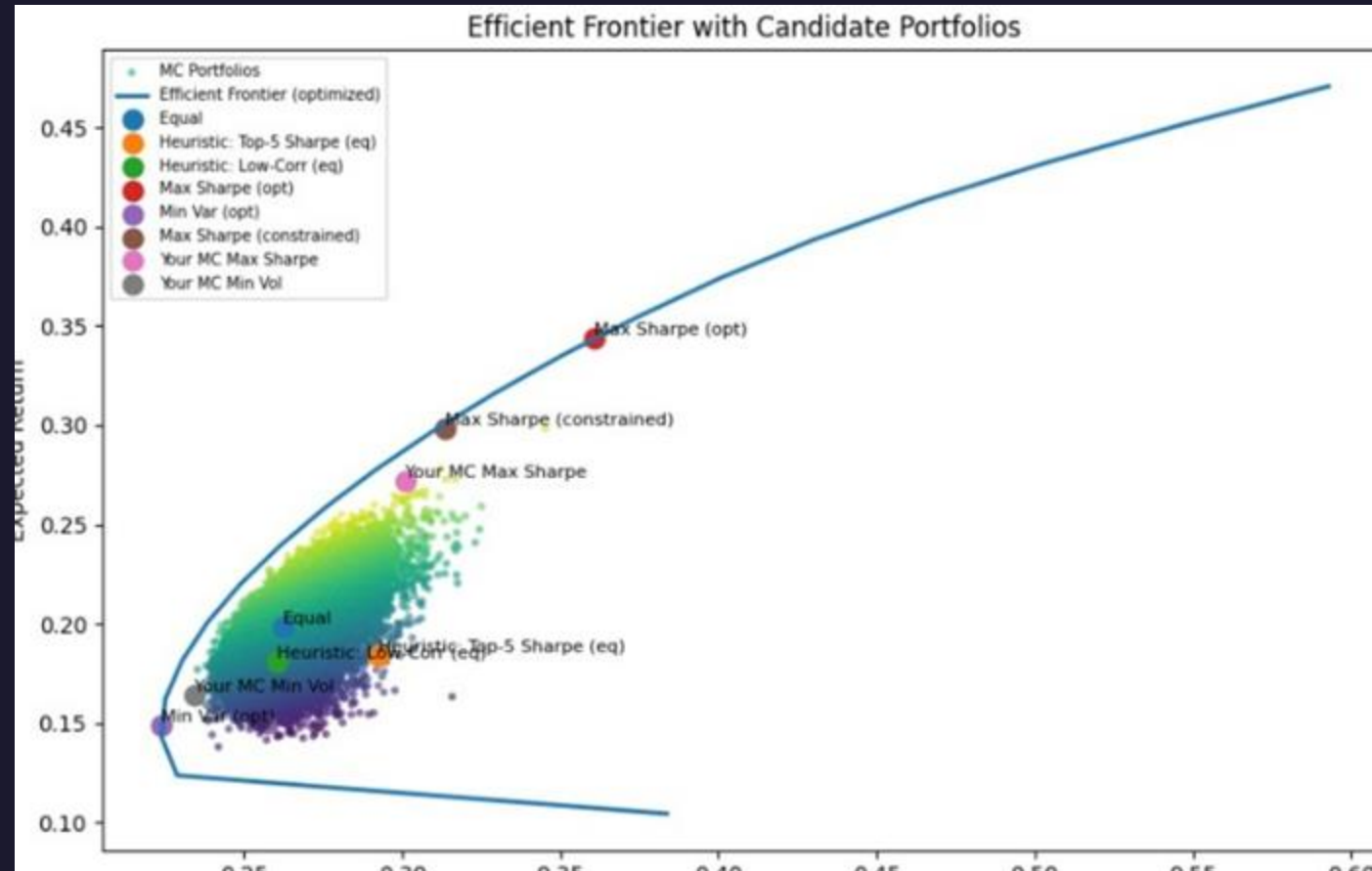
H1: A diversified portfolio across the three sectors has a significantly higher Sharpe Ratio than a single-sector portfolio.



Insight: A diversified portfolio across the three sectors has a significantly higher Sharpe Ratio than a single-sector portfolio

Diversified vs. Sector-Only Performance (Test Period 2022–2025)

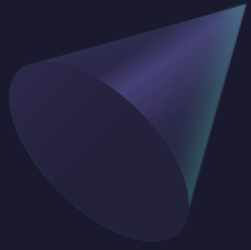
- **Efficient Frontier:** Curve showing the optimal portfolios that maximize expected return for each level of risk.
- **Monte Carlo Simulation:** Thousands of random portfolios plotted as dots; most lie below the frontier, indicating inefficiency.
- **Equal-Weight Portfolio:** Positioned near the center, offering balanced but modest risk and return.
- **Minimum-Variance Portfolio:** Located at the far left, delivering the lowest risk but also lower return.
- **Maximum Sharpe Portfolio:** Sits directly on the frontier, achieving the best risk-adjusted return.



Hypothesis 2: Diversification Benefits

- **Optimized Portfolios (Red & Purple):** Clearly outperform naive and heuristic approaches.
- **Key Insight:** Proper diversification and optimization enable portfolios to align with the efficient frontier.
- **Result:** Maximized efficiency through superior risk–return trade-offs.
- **Diversified Min-Variance Portfolio** outperforms Max-Sharpe, Equal-Weight, and single-sector portfolios (Sharpe Ratio = **1.286**).
- **Diversification Effect:** Combining energy, technology, and finance reduces volatility and improves risk-adjusted returns.
- **Conclusion:** Reject H_0 ; diversification across sectors **significantly enhances portfolio performance**.





Hypothesis 3: Predictive Model Effectiveness

Hypothesis: Using predictive models can improve the accuracy of forecasting CVX daily returns compared to a naive “no-prediction” baseline.

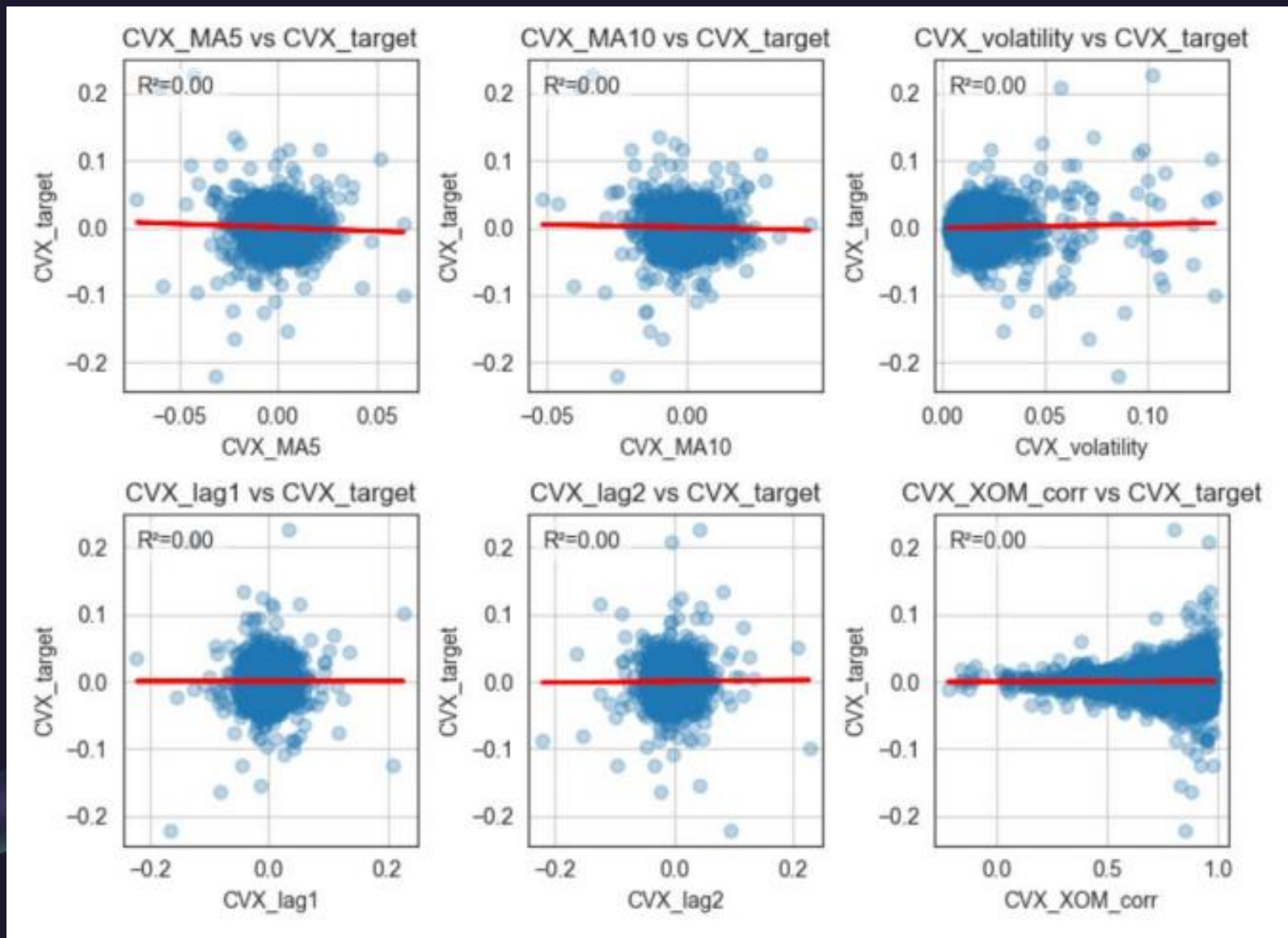
Null Hypothesis (H0): The predictive model does not produce forecasts that are significantly more accurate than a naive strategy (e.g., predicting tomorrow’s return as today’s return).

Alternative Hypothesis (H1): The predictive model produces forecasts that are significantly more accurate than a naive strategy.



Hypothesis 3: Predictive Model Effectiveness

Linearity
validation



Hypothesis 3: Predictive Model Effectiveness

XGBOOST

Test MSE:

0.000332

Test R^2 : -0.1346

```
# Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, shuffle=False)

# Initialize XGBoost regressor
model = xgb.XGBRegressor(
    ... n_estimators=500,
    ... max_depth=2,
    ... learning_rate=0.05,
    ... objective='reg:squarederror',
    ... random_state=42
)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

Hypothesis 3: Predictive Model Effectiveness

Random Forest

Test MSE:

0.000292

Test R^2 : 0.0014

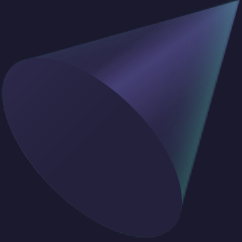
```
# Split into train and test sets (no shuffle since it's time series)
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, shuffle=False)

# Initialize Random Forest Regressor
rf_model = RandomForestRegressor(
    n_estimators=500,
    max_depth=6,
    random_state=42
)

# Train model
rf_model.fit(X_train, y_train)

# Predict on test set
y_pred = rf_model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```



Hypothesis 3: Predictive Model Effectiveness



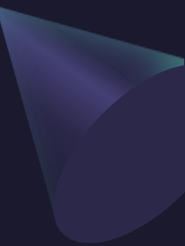
Therefore, the null hypothesis (H_0) of no improvement over a naive strategy can be rejected in favor of H_1 , which aligns with the predictive insights observed from the model evaluations.

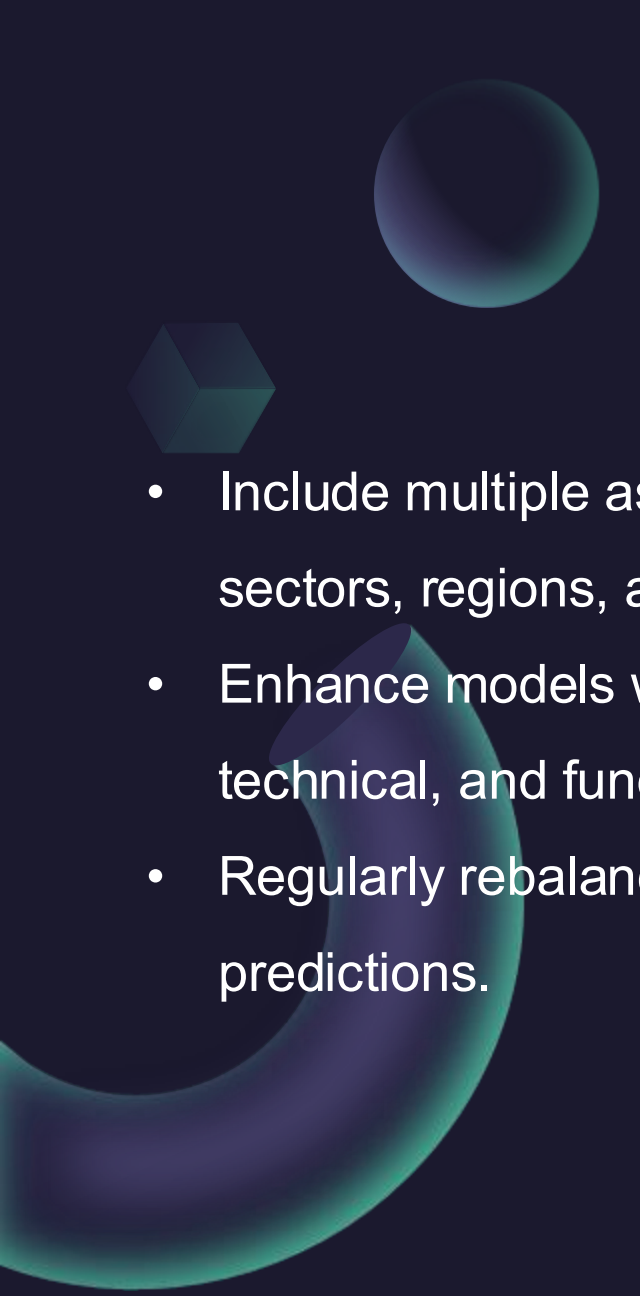




Conclusions

- This reinforces a key principle in finance: **you cannot predict markets reliably, but you can manage risk intelligently through diversification and optimization.**
- **Risk-Return Tradeoff:** Technology offers high returns but is very volatile. Finance and Energy are steadier but lower return.
- **Diversification:** Combining sectors significantly improves performance because of low inter-sector correlations.
- **Optimization:** Data-driven optimized portfolios outperformed naive approaches, proving the strength of systematic allocation.
- **Predictive Models:** Our models failed to beat even simple strategies, suggesting that forecasting daily returns is far less effective than diversification and optimization.





Recommendations

- Include multiple asset classes (stocks, bonds, REITs, commodities) and diversify across sectors, regions, and market caps.
- Enhance models with LSTM, XGBoost, or ensembles, including macroeconomic, technical, and fundamental features.
- Regularly rebalance, backtest, and stress-test the portfolio to manage risk and validate predictions.